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ALU DATA PATH

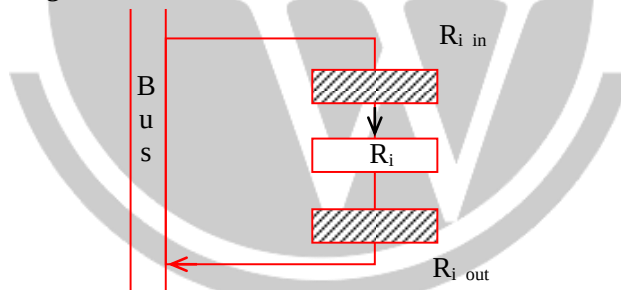
3.1 Introduction

- The sequence of operations involved in processing an instruction constitutes an instruction cycle. The instruction cycle consists of phases like.
 - (1) Fetch cycle
 - (2) Decode cycle
 - (3) Operand fetch cycle
 - (4) Execute cycle
- To perform these, the processor unit has to perform set of operations called “Micro-Operations”.

3.2 The Basic Operations Performed are

3.2.1 Register transfers:

- In general, the input & output of register R_i are connected to the bus via switched controlled by the signals R_i in and R_i out



- When R_i in is set to 1, the data on the bus are loaded into R_i .
- When R_i out is set to 1, the contents of register R_i are placed on the bus.
- To transfer the contents of R_1 to register R_4 : $R_4 \leftarrow R_1$.
- Enable the output of register R_1 by setting R_1 out to 1. This places the contents of R_1 on the processor bus.
- Enable the input to register R_4 by setting R_4 in to 1. This loads data from the processor bus into register R_4 .

3.2.2 Performing ALU operations

The ALU has two inputs A and B and one output

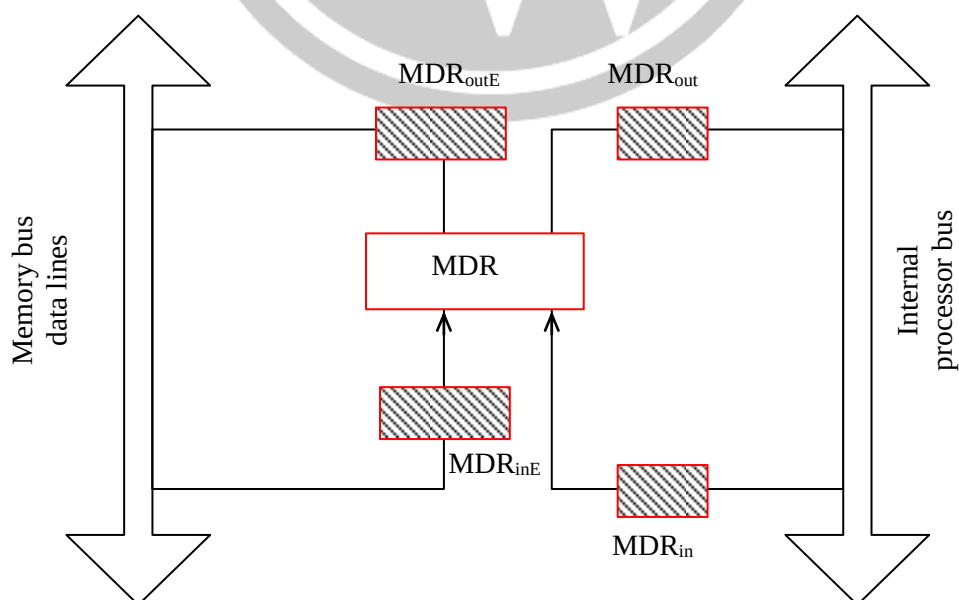
- A gets operand from output of MUX
- B gets operand from bus
- The result is gets stored in Z-register.

The sequence of steps for ALU operation $R_3 \leftarrow R_1$

- (1) The contents of R_1 are loaded in Y .
 - (2) The contents of $Y_1 R_2$ are applied to A & B inputs of ALU, performs ALU operation & stores the result in Z-register.
 - (3) The contents of Z are stored in R_3 .
- The sequence of operations is
 - (1) R_1 out, Y in
 - (2) R_2 out, select Y, Add, Z in
 - (3) Z out, R_3 in
 - The functions performed by ALU depends on the signals applied to its control lines. (Here Add line is set to 1).
 - Only one register output can be connected to the bus during any clock cycle.
 - The no of steps indicates no of clocks.

3.2.3 Fetching a word from memory (read operation)

- To read a memory word, consider MOV (R_1), R_2 . The action needed to execute this instruction are
 - (1) $MAR \leftarrow [R_1]$
 - (2) Start a read operation on the memory bus
 - (3) Wait for the MFC (Memory function to complete) response from memory.
 - (4) Load MDR from the memory bus
 - (5) $R_2 \leftarrow [MDR]$
- Hence the memory read operation sequence of operations is
 - (1) R_1 out, MAR in, Read
 - (2) WMFC (wait for memory operation to complete), MDR inE
 - (3) MDR out, R_2 in



3.2.4 Storing a word in Memory (write operation)

- The sequence of steps for write operation MOV R₂,
 - (1) The desired address is loaded into MAR
 - (2) The data to be written is loaded into MDR
 - (3) Write command is issued.
 - (4) Wait for memory operation to complete.
- The sequence of operations is
 - (1) R_{1 out}, MAR_{in}
 - (2) R_{2out}, MDR_{in}, write
 - (3) MDR_{outE}, WMFC

3.2.5 Branch Instructions

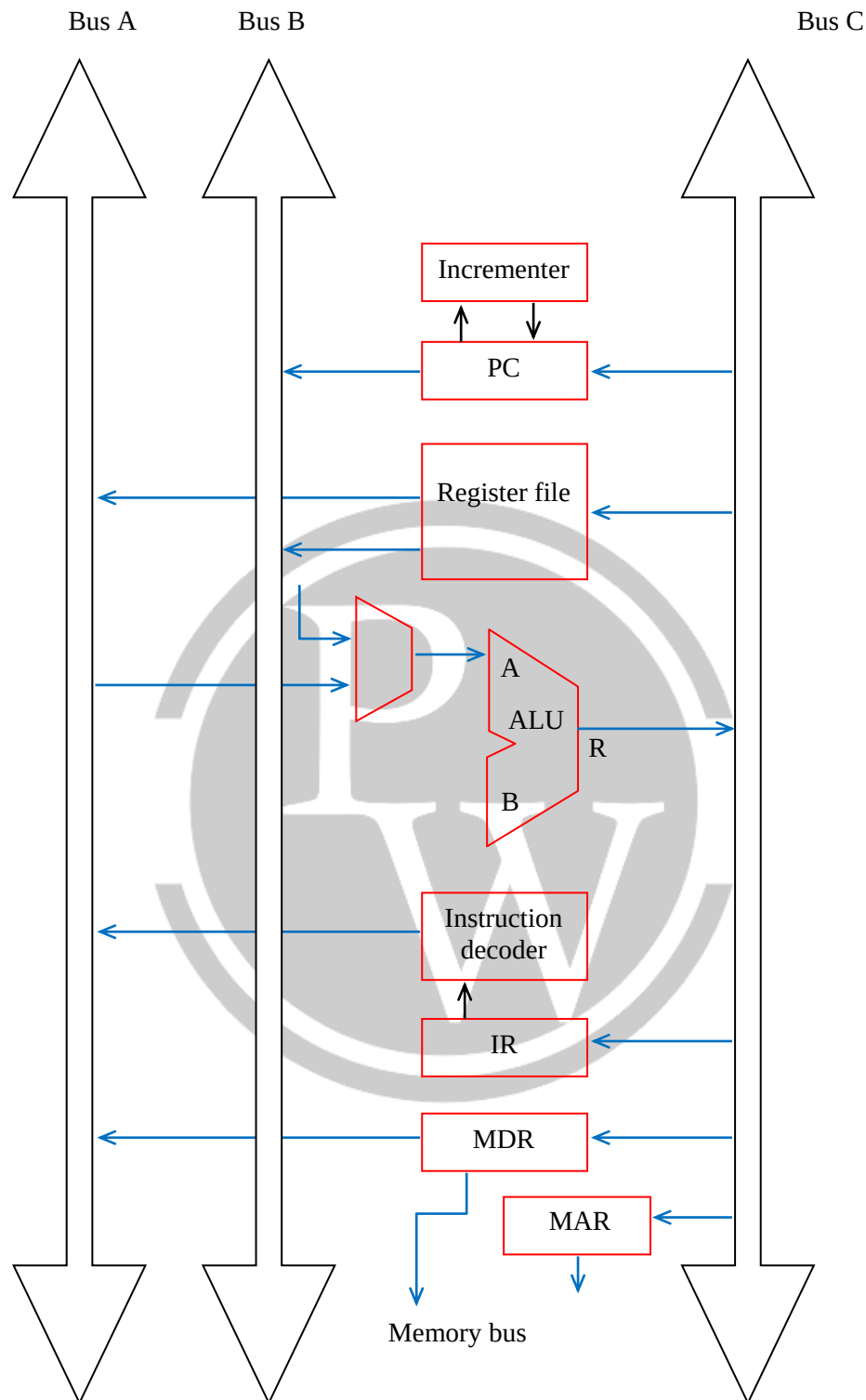
- A branch instruction replaces the content of PC with the branch target address. The address is obtained by adding an offset X, which is given in the branch instruction, to the updated value of PC.
- The control sequence for branching (unconditional is
 - (1) PC_{out}, PC_{in}, Y_{in}, WMFC
 - (2) Z_{out}, PC_{in}, Y_{in}, WMFC
 - (3) MDR_{out}, IR_{in}
 - (4) Offset – of – IR_{out}, Add, Z_{in}

3.2.6 Execution of complete Instruction

- Executing an instruction requires (Add (R₃), R₁)
 - (1) Fetch the instruction
 - (2) Fetch the first operand (memory location specified by R₃)
 - (3) Perform the addition
 - (4) Load the result into R₁
- The control sequence for the execution of ADD (R₃), R₁ in a single-bus organization is
 - (1) PC_{out}, MAR_{in}, Read, Select, Add, Z_{in}
 - (2) Z_{out}, PC_{in}, Y_{in}, WMFC
 - (3) MDR_{out}, IR_{in}
 - (4) R_{3out}, MAR_{in}, Read
 - (5) R_{1 out}, Y_{in}, WMFC
 - (6) MDR_{out}, select Y, Add, Z_{in}
 - (7) Z_{out}, R_{1 in}, End

3.3 Multiple – Bus Organization

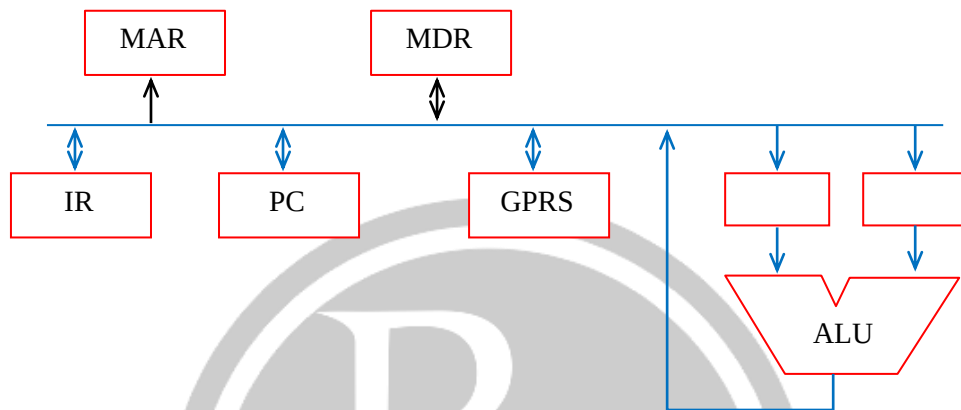
- With simple single bus structure, the resulting control sequence is quite long because only one data item can be transferred over the bus in a clock cycle. To reduce the number of steps needed, most commercial processors provide multiple internal paths that enable several transfers to take place in parallel.
- The three bus organization of data path is



- All general – purpose registers are combined into a single block called register file. It has two – outputs, allowing the contents of two different registers to be accessed simultaneously and have their contents placed on buses A and B. The data on bus ‘C’ to be loaded into third register during same clock.

- Buses A and B are used to transfer the source operands to the A and B inputs of ALU, the result is transferred to the destination over bus C.
- The control sequence for instruction ADD R₄, R₅, R₆
 - (1) P_{Cout}, R = B, MAR_{in}, Read, Inc PC
 - (2) WMFC
 - (3) MDT_{out} B, R = B, IR_{in}
 - (4) R_{4 outA}, R_{5 outB}, select A, Add, R_{6 in}, end.

Example (1)



The ALU, bus & registers in data path are of identical size. All operations including incrementing of PC and GPRSs are to be carried out in the ALU. 2-clock cycles are needed for memory read operation. (one for loading address in MAR and loading data from memory into the MDR).

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